## EXERCISE-I (Conceptual Questions) — Build Up Your Understanding

### OXIDATION NUMBER

**1.** In $[Ni(CO)_4]$, the oxidation state of Ni is :
(1) 4 (2) 0 (3) 2 (4) 8
**RR0001**

**2.** The oxidation number of nitrogen in $NH_2OH$ is :
(1) 0 (2) +1 (3) –1 (4) –2
**RR0002**

**3.** Of the following elements, which one has the same oxidation state in all of its compounds ?
(1) Hydrogen (2) Fluorine
(3) Carbon (4) Oxygen
**RR0003**

**4.** Oxidation number of fluorine in $OF_2$ is :
(1) +1 (2) +2 (3) –1 (4) –2
**RR0004**

**5.** The oxidation number of C in $CH_4$, $CH_3Cl$, $CH_2Cl_2$, $CHCl_3$ and $CCl_4$ are respectively :
(1) +4, +2, 0, –2, –4 (2) +2, +4, 0, –4, –2
(3) –4, –2, 0, +2, +4 (4) –2, –4, 0, +4,. +2
**RR0005**

**6.** Phosphorus has the oxidation state of +3 in :
(1) Ortho phosphoric acid (2) Phosphorus acid
(3) Meta phosphoric acid (4) Pyrophosphoric acid
**RR0006**

**7.** Oxidation state of oxygen in hydrogen peroxide is
(1) –1 (2) +1 (3) 0 (4) –2
**RR0007**

**8.** Which one of the following statements is not correct?
(1) Oxidation state of S in $(NH_4)_2S_2O_8$ is +6
(2) Oxidation number of Os in $OsO_4$ is +8
(3) Oxidation state of S in $H_2SO_5$ is +8
(4) Oxidation number of O in $KO_2$ is $-\frac{1}{2}$
**RR0009**

**9.** Which of the following shows highest oxidation number in combined state :
(1) Os
(2) Ru
(3) Both (1) and (2)
(4) None
**RR0010**

**10.** Oxidation number of sodium in sodium amalgam(Na-Hg) is :
(1) +2 (2) +1 (3) –3 (4) Zero
**RR0011**

**11.** Oxidation number of C in HNC is :
(1) +2 (2) –3 (3) +3 (4) Zero
**RR0013**

**12.** Oxidation number of Fe in $Fe_{0.94}O$ is :
(1) 200 (2) 200/94
(3) 94/200 (4) None
**RR0014**

**13.** Oxidation number of carbon in carbon suboxide ($C_3O_2$) is :
(1) $\frac{+2}{3}$ (2) $\frac{+4}{3}$ (3) +4 (4) $\frac{-4}{3}$
**RR0015**

**14.** Oxidation number of sulphur in $Na_2S_2O_3$ would be :-
(1) +2 (2) +4 (3) –2 (4) 0
**RR0016**

**15.** Two oxidation states for chlorine are found in the compound :
(1) $CaOCl_2$ (2) KCl (3) $KClO_3$ (4) $Cl_2O_7$
**RR0017**

**16.** –1/3 oxidation state of nitrogen will be obtained in case of :
(1) Ammonia ($NH_3$) (2) Hydrazoic acid ($N_3H$)
(3) Nitric oxide (NO) (4) Nitrous oxide ($N_2O$)
**RR0019**

**17.** Compound $YBa_2Cu_3O_7$ is a super conductor. The O.N. of the copper in the compound will be:[O.No. of Y=+3]
(1) +7/3 (2) zero (3) +2 (4) +1
**RR0021**

**18.** The oxidation state of iodine in $H_4IO_6^-$ is :-
(1) +7 (2) –1 (3) +5 (4) +1
**RR0022**

**19.** Amongst the following, identify the species with an atom in + 6 oxidation state:-
(1) $MnO_4^-$ (2) $Cr(CN)_6^{3-}$
(3) $NiF_6^{2-}$ (4) $CrO_2Cl_2$
**RR0023**

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**20.** The oxidation state of + 1 for phosphorous is found in:-
(1) Phosphorous acid ($H_3PO_3$)
(2) Orthophosphoric acid ($H_3PO_4$)
(3) Hypo phosphorous acid ($H_3PO_2$)
(4) Hypo phosphoric acid ($H_4P_2O_6$)

**RR0024**

**21.** In which of the following compounds iron has lowest oxidation state:-
(1) $FeSO_4(NH_4)_2SO_4.6H_2O$
(2) $K_4[Fe(CN)_6]$
(3) $[Fe(CO)_5]$
(4) $Fe_{0.94}O$

**RR0025**

**22.** Select the compound in which the oxidation number of oxygen is –1:-
(1) $H_2O$ (2) $O_2F_2$
(3) $Na_2O$ (4) $BaO_2$

**RR0026**

**23.** Match List - I (compound) with list - II (Oxidation state of N) and select the correct answer using the codes given below the list:-

| List - I | List-II |
|---|---|
| (A) $KNO_3$ | (a) – 1/3 |
| (B) $HNO_2$ | (b) – 3 |
| (C) $NH_4Cl$ | (c) 0 |
| (D) $NaN_3$ | (d) + 3 |
| | (e) + 5 |

Codes are:-

| | A | B | C | D |
|---|---|---|---|---|
| (1) | e | d | b | a |
| (2) | e | b | d | a |
| (3) | d | e | a | c |
| (4) | b | c | d | e |

**RR0027**

**24.** In which of the following pair, oxidation number of Fe is same :-
(1) $K_3[Fe(CN)_6]$, $Fe_2O_3$ (2) $Fe(CO)_5$, $Fe_2O_3$
(3) $Fe_2O_3$, FeO (4)$Fe_2(SO_4)_3$, $K_4[Fe(CN)_6]$

**RR0028**

**25.** In the conversion of $Br_2$ to $BrO_3^-$, the oxidation state of bromine changes from :-
(1) 0 to 5 (2) 1 to 5
(3) 0 to –3 (4) 2 to 5

**RR0029**

**26.** The sum of oxidation states of sulphur in $H_2S_2O_8$ is :-
(1) +2 (2) +6 (3) +7 (4) +12

**RR0030**

**27.** In which of the following compounds of Cr, the oxidation number of Cr is not +6 :-
(1) $CrO_3$ (2) $CrO_2Cl_2$
(3) $Cr_2O_3$ (4) $K_2Cr_2O_7$

**RR0031**

**28.** Oxidation state of cobalt in $[Co(NH_3)_4(H_2O)Cl]SO_4$ is
(1) 0 (2) +4 (3) –2 (4) +3

**RR0032**

**29.** Oxidation number of carbon in graphite is :-
(1) Zero (2) +1 (3) +4 (4) +2

**RR0033**

**30.** Oxidation number of 'N' in $N_3H$ (hydrazoic acid) is :-
(1) $-\frac{1}{3}$ (2) –3 (3) +3 (4) $+\frac{2}{3}$

**RR0034**

**31.** Which of the following doesn't have +5 oxidation state of phosphorus ?
(1) Phosphorus acid ($H_3PO_3$)
(2) Orthophosphoric acid ($H_3PO_4$)
(3) Meta phosphoric acid ($HPO_3$)
(4) Pyro phosphoric acid ($H_4P_2O_7$)

**RR0035**

**32.** The oxidation number of arsenic atom in $H_3AsO_4$ is :-
(1) –1 (2) –3 (3) +3 (4) +5

**RR0036**

**33.** In substance $Mg(HXO_3)$, the oxidation number of X is :-
(1) 0 (2) +2 (3) +3 (4) +4

**RR0037**

**34.** The oxidation number of phosphorus in $PH_4^+$, $PO_2^{3-}$, $PO_4^{3-}$ and $PO_3^{3-}$ are respectively :–
(1) –3, +1, +3, +5 (2) –3, +3, +5, +1
(3) +3, –3, +5, +1 (4) –3, +1, +5, +3

**RR0040**

**35.** Which of the following compounds are arranged in increasing oxidation number of S :–
(1) $H_2SO_3$, $H_2S$, $H_2SO_4$, $H_2S_2O_3$
(2) $H_2S_2O_3$, $H_2SO_3$, $H_2S$, $H_2SO_4$
(3) $H_2S$, $H_2SO_3$, $H_2SO_4$, $H_2S_2O_3$
(4) $H_2S$, $H_2S_2O_3$, $H_2SO_3$, $H_2SO_4$

**RR0041**

**36.** Iodine shows the highest oxidation state in the compound :–
(1) KI (2) $KI_3$ (3) $IF_5$ (4) $KIO_4$

**RR0042**

**37.** The sum of the oxidation states of all the carbon atoms present in the compound $C_6H_5CHO$ is :
(1) –4 (2) 3 (3) + 5 (4) – 4/7

**RR0043**

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## APPLICATIONS OF REDOX REACTIONS

**38.** A reducing agent is a substance which can :
(1) Accept electrons
(2) Donate electrons
(3) Accept protons
(4) Donate protons

**RR0045**

**39.** The reaction $H_2S + H_2O_2 \rightarrow S + 2H_2O$ manifests :
(1) Oxidising action of $H_2O_2$
(2) Reducing nature of $H_2O_2$
(3) Acidic nature of $H_2O_2$
(4) Alkaline nature of $H_2O_2$

**RR0046**

**40.** If an element is in its lowest oxidation state, under proper conditions it can act as :
(1) Reducing agent
(2) An oxidising agent
(3) Oxidising as well as reducing agent
(4) Neither oxidising nor reducing agent

**RR0047**

**41.** In a reaction of $H_2O$ (steam) + C (glowing) $\rightarrow$ CO + $H_2$
(1) $H_2O$ is the reducing agent
(2) $H_2O$ is the oxidising agent
(3) carbon is the oxidising agent
(4) oxidation-reduction does not occur

**RR0048**

**42.** The compound that can work both as an oxidising as well as reducing agent is :
(1) $KMnO_4$ (2) $H_2O_2$
(3) $Fe_2(SO_4)_3$ (4) $K_2Cr_2O_7$

**RR0049**

**43.** $HNO_2$ acts as an oxidant with which one of the following reagent:-
(1) $KMnO_4$ (2) $H_2S$ (3) $K_2Cr_2O_7$ (4) $Br_2$

**RR0051**

**44.** In which of the following reaction $H_2O_2$ acts as reducing agent :-
(1) $2FeCl_2 + 2HCl + H_2O_2 \rightarrow 2FeCl_3 + 2H_2O$
(2) $Cl_2 + H_2O_2 \rightarrow 2HCl + O_2$
(3) $2HI + H_2O_2 \rightarrow 2H_2O + I_2$
(4) $H_2SO_3 + H_2O_2 \rightarrow H_2SO_4 + H_2O$

**RR0052**

**45.** When $H_2$ reacts with Na, it acts as :-
(1) Oxidising agent (2) Reducing agent
(3) Both (4) Cannot be predicted

**RR0054**

**46.** In the course of a chemical reaction an oxidant –
(1) Loses electron
(2) Gains electron
(3) Both loses and gain electrons
(4) Electron change does not occur

**RR0056**

**47.** In the reaction:-
$C + 4HNO_3 \rightarrow CO_2 + 2H_2O + 4NO_2$
$HNO_3$ acts as :–
(1) An oxidising agent (2) An acid
(3) A reducing agent (4) A base

**RR0057**

**48.** A compound contains atoms A, B and C. The oxidation number of A is +2, of B is +5 and of C is –2. The possible formula of the compound is :
(1) $ABC_2$ (2) $B_2(AC_3)_2$
(3) $A_3(BC_4)_2$ (4) $A_3(B_4C)_2$

**RR0058**

**49.** Equivalent weight of $N_2$ in the change $N_2 \rightarrow NH_3$ is
(1) $\frac{28}{6}$ (2) 28 (3) $\frac{28}{2}$ (4) $\frac{28}{3}$

**RR0059**

**50.** Equivalent weight of $NH_3$ in the change $N_2 \rightarrow NH_3$ is :
(1) $\frac{17}{6}$ (2) 17 (3) $\frac{17}{2}$ (4) $\frac{17}{3}$

**RR0060**

**51.** In the reaction, $2S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$, the eq. wt. of $Na_2S_2O_3$ is equal to its :
(1) Mol. wt. (2) Mol. wt./2
(3) 2 x Mol. wt. (4) Mol. wt./6

**RR0061**

**52.** In the reaction, $VO + Fe_2O_3 \rightarrow FeO + V_2O_5$, the eq. wt. of $V_2O_5$ is equal to its :
(1) Mol. wt. (2) Mol. wt./8
(3) Mol .wt./6 (4) Mol. wt./2

**RR0062**

**53.** Molecular weight of $KBrO_3$ is M. What is its equivalent weight, if the reaction is :
$BrO_3^- \rightarrow Br^-$ (acidic medium)
(1) M (2) M/4 (3) M/6 (4) 6M

**RR0064**

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**54.** In the reaction : $A^{-n_2} + xe^- \rightarrow A^{-n_1}$, here x will be

(1) $n_1 + n_2$ (2) $n_2 - n_1$

(3) $n_1 - n_2$ (4) $n_1 . n_2$

**RR0065**

**55.** What would be the equivalent weight of the reductant in the reaction :

$[Fe(CN)_6]^{-3} + H_2O_2 + 2OH^- \rightarrow 2[Fe(CN)_6]^{-4} + 2H_2O + O_2$

[Given : Fe = 56, C = 12, N = 14, O = 16, H = 1]

(1) 17 (2) 212 (3) 34 (4) 32

**RR0066**

**56.** Equivalent weight of $FeC_2O_4$ in the change : $FeC_2O_4 \rightarrow Fe^{3+} + CO_2$ is :

(1) M/3 (2) M/6

(3) M/2 (4) M/1

**RR0068**

**57.** The number of mole of oxalate ions oxidised by one mole of $MnO_4^-$ is :

(1) 1/5 (2) 2/5

(3) 5/2 (4) 5

**RR0070**

**58.** In a reaction 4 mole of electrons are transferred to one mole of $HNO_3$ when it acts as an oxidant. The possible reduction product is :

(1) (1/2) mole $N_2$ (2) (1/2) mole $N_2O$

(3) 1 mole of $NO_2$ (4) 1 mole $NH_3$

**RR0072**

**59.** The equivalent weight of $MnSO_4$ is half of its molecular weight when it is converted to :-

(1) $Mn_2O_3$ (2) $MnO_2$

(3) $MnO_4^-$ (4) $MnO_4^{-2}$

**RR0073**

**60.** In the following change, $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$
If the atomic weight of iron is 56, then its equivalent weight will be :-

(1) 42 (2) 21 (3) 63 (4) 84

**RR0074**

**61.** $Cr_2O_7^{-2} + I^- + H^+ \rightarrow Cr^{+3} + I_2 + H_2O$
The equivalent weight of the reductant in the above equation is :– (At. wt. of Cr=52, I=127)

(1) 26 (2) 127 (3) 63.5 (4) 10.4

**RR0075**

**62.** How many moles of $KMnO_4$ are reduced by 1 mole of ferrous oxalate in acidic medium:-

(1) $\frac{1}{5}$ (2) $\frac{5}{3}$ (3) $\frac{1}{3}$ (4) $\frac{3}{5}$

**RR0076**

## REDOX REACTIONS

**63.** Which one of the following is a redox reaction ?

(1) $H_2 + Br_2 \rightarrow 2HBr$

(2) $2NaCl + H_2SO_4 \rightarrow Na_2SO_4 + 2HCl$

(3) $HCl + AgNO_3 \rightarrow AgCl + HNO_3$

(4) $NaOH + HCl \rightarrow NaCl + H_2O$

**RR0078**

**64.** Which of the following is not a redox change ?

(1) $2H_2S + SO_2 \rightarrow 2H_2O + 3S$

(2) $2BaO + O_2 \rightarrow 2BaO_2$

(3) $BaO_2 + H_2SO_4 \rightarrow BaSO_4 + H_2O_2$

(4) $2KClO_3 \rightarrow 2KCl + 3O_2$

**RR0079**

**65.** In the reaction, $Cl_2 + OH^- \rightarrow Cl^- + ClO_4^- + H_2O$, chlorine is :

(1) Oxidised

(2) Reduced

(3) Oxidised as well as reduced

(4) Neither oxidised nor reduced

**RR0081**

**66.** Which is a redox reaction :

(1) $2CuI_2 \rightarrow CuI + I_2$

(2) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$

(3) $NH_4Cl + NaOH \rightarrow NH_3 + NaCl + H_2O$

(4) $Cr_2(SO_4)_3 + 6KOH \rightarrow 2Cr(OH)_3 + 3K_2SO_4$

**RR0082**

**67.** Which of the following example does not represent disproportionation -

(1) $MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$

(2) $2H_2O_2 \rightarrow 2H_2O + O_2$

(3) $4KClO_3 \rightarrow 3KClO_4 + KCl$

(4) $3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$

**RR0083**

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**68.** The decomposition of $KClO_3$ to KCl and $O_2$ on heating is an example of :
(1) Intermolecular redox change
(2) Intramolecular redox change
(3) Disproportionation or auto redox change
(4) Comproportionation

**RR0084**

**69.** Which of the following change represents a disproportionation reaction (s) :
(1) $Cl_2 + 2OH^- \rightarrow ClO^- + Cl^- + H_2O$
(2) $Cu_2O + 2H^+ \rightarrow Cu + Cu^{2+} + H_2O$
(3) $2HCuCl_2 \xrightarrow[\text{Water}]{\text{dilution with}} Cu + Cu^{2+} + 4Cl^- + 2H^+$
(4) All of the above

**RR0085**

**70.** How many electrons should $X_2H_4$ liberate so that in the new compound X shows oxidation number of -½ (E.N. X > H)
(1) 10 (2) 4 (3) 3 (4) 2

**RR0087**

**71.** Which one of the following is not a redox reaction :-
(1) $CaCO_3 \rightarrow CaO + CO_2$
(2) $2H_2 + O_2 \rightarrow 2H_2O$
(3) $Na + H_2O \rightarrow NaOH + \frac{1}{2}H_2$
(4) $MnCl_3 \rightarrow MnCl_2 + \frac{1}{2}Cl_2$

**RR0088**

**72.** In the reaction -

$MnO_4^- + SO_3^{2-} + H^+ \rightarrow SO_4^{-2} + Mn^{+2} + H_2O$

(1) $MnO_4^-$ and $H^+$ both are reduced
(2) $MnO_4^-$ is reduced and $H^+$ is oxidised
(3) $MnO_4^-$ is reduced and $SO_3^{2-}$ is oxidised
(4) $MnO_4^-$ is oxidised and $SO_3^{2-}$ is reduced

**RR0089**

**73.** $I_2 + KI \rightarrow KI_3$
In the above reaction:-
(1) Only oxidation taken place
(2) Only reduction takes place
(3) Both the above
(4) Neither oxidation nor reduction

**RR0090**

**74.** Which of the following reaction represents the oxidising behaviour of $H_2SO_4$ :-
(1) $2PCl_5 + H_2SO_4 \rightarrow 2POCl_3 + 2HCl + SO_2Cl_2$
(2) $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$
(3) $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$
(4) $2HI + H_2SO_4 \rightarrow I_2 + SO_2 + 2H_2O$

**RR0091**

**75.** Select the example of disproportionation reaction
(1) $BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + 2HCl$
(2) $NH_4NO_3 \rightarrow N_2O + 2H_2O$
(3) $4H_3PO_3 \rightarrow PH_3 + 3H_3PO_4$
(4) $AgCl + 2NH_3 \rightarrow Ag(NH_3)_2Cl$

**RR0092**

**76.** Which of the following reaction involves oxidation & reduction :-
(1) $NaBr + HCl \rightarrow NaCl + HBr$
(2) $HBr + AgNO_3 \rightarrow AgBr + HNO_3$
(3) $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$
(4) $H_2 + Br_2 \rightarrow 2HBr$

**RR0093**

**77.** The reaction
$2K_2MnO_4 + Cl_2 \rightarrow 2KMnO_4 + 2KCl$ is an example of
(1) Redox
(2) Reduction only
(3) Neutralization
(4) Disproportionation

**RR0094**

**78.** Which of the following reaction involves neither oxidation nor reduction :–
(1) $CrO_4^{2-} \rightarrow Cr_2O_7^{2-}$
(2) $Cr \rightarrow CrCl_3$
(3) $Na \rightarrow Na^+$
(4) $2S_2O_3^{2-} \rightarrow S_4O_6^{2-}$

**RR0095**

**79.** $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
Zn undergoes –
(1) Reduction
(2) Oxidation
(3) Both oxidation and reduction
(4) Neither oxidation nor reduction

**RR0096**

## BALANCING OF REDOX REACTIONS

**80.** Balance the following given half reaction for the unbalanced whole reaction :
$CrO_4^{2-} \rightarrow CrO_2^- + OH^-$ is :
(1) $CrO_4^{-2} + 2H_2O + 3e^- \rightarrow CrO_2^- + 4OH^-$
(2) $2CrO_4^{-2} + 8H_2O \rightarrow CrO_2^- + 4H_2O + 8OH^-$
(3) $CrO_4^{-2} + H_2O \rightarrow CrO_2^- + H_2O + OH^-$
(4) $3CrO_4^{-2} + 4H_2O + 6\,e^- \rightarrow 2CrO_2^{-1} + 8OH^-$

**RR0097**

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**81.** Choose the set of coefficients that correctly balances the following equation :

$x\ Cr_2O_7^{2-} + yH^+ + z\ e^- \rightarrow a\ Cr^{+3} + bH_2O$

| | x | y | z | a | b |
|---|---|---|---|---|---|
| (1) | 2 | 14 | 6 | 2 | 7 |
| (2) | 1 | 14 | 6 | 2 | 7 |
| (3) | 2 | 7 | 6 | 2 | 7 |
| (4) | 2 | 7 | 6 | 1 | 7 |

**RR0098**

**82.** In the reaction: $MnO_4^- + xH^+ + ne^- \rightarrow Mn^{2+} + yH_2O$ What is the value of n :

(1) 5 (2) 8 (3) 6 (4) 3

**RR0099**

**83.** The number of electrons required to balance charge in the following equation –

$MnO_4^- + 2H_2O \longrightarrow MnO_2 + 4OH^-$ is

(1) 5 (2) 4 (3) 3 (4) 2

**RR0100**

**84.** $2KMnO_4 + 5H_2S + 6H^+ \rightarrow 2Mn^{2+} + 2K^+ + 5S + 8H_2O$. In the above reaction, how many electrons would be involved in the oxidation of 1 mole of reductant?

(1) Two (2) Five (3) Ten (4) One

**RR0102**

**85.** The value of *n* in : $MnO_4^- + 8H^+ + ne^- \rightarrow Mn^{2+} + 4H_2O$ is

(1) 5 (2) 4 (3) 3 (4) 2

**RR0103**

**86.** For the redox reaction

$Zn + NO_3^- \rightarrow Zn^{2+} + NH_4^+$ in basic medium, coefficients of Zn, $NO_3^-$ and $OH^-$ in the balanced equation respectively are :

(1) 4, 1, 7 s (2) 7, 4, 1

(3) 4, 1, 10 (4) 1, 4, 10

**RR0105**

**87.** In the balanced equation-

$[Zn + H^+ + NO_3^- \rightarrow NH_4^+ + Zn^{+2} + H_2O]$ coefficient of $NH_4^+$ is:-

(1) 4 (2) 3 (3) 2 (4) 1

**RR0106**

**88.** In the balanced equation

$MnO_4^- + H^+ + C_2O_4^{2-} \rightarrow Mn^{2+} + CO_2 + H_2O$, the moles of $CO_2$ formed are :-

(1) 2 (2) 4 (3) 5 (4) 10

**RR0107**

**89.** In the following reaction the value of 'X' is

$H_2O + SO_3^{2-} \rightarrow SO_4^{2-} + 2H^+ + X$

(1) $4e^-$ (2) $3e^-$ (3) $2e^-$ (4) $1e^-$

**RR0108**

**90.** The number of electrons required to balance the following equation are:

$NO_3^- + 4H^+ \rightarrow 2H_2O + NO$

(1) 2 on right side (2) 3 on left side

(3) 3 on right side (4) 5 on left side

**RR0109**

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## EXERCISE-I (Conceptual Questions) ANSWER KEY

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Ans.** | **2** | **3** | **2** | **3** | **3** | **2** | **1** | **3** | **3** | **4** | **1** | **2** | **2** | **1** | **1** |
| Que. | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| **Ans.** | **2** | **1** | **1** | **4** | **3** | **3** | **4** | **1** | **1** | **1** | **4** | **3** | **4** | **1** | **1** |
| Que. | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 |
| **Ans.** | **1** | **4** | **3** | **4** | **4** | **4** | **1** | **2** | **1** | **1** | **2** | **2** | **2** | **2** | **1** |
| Que. | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| **Ans.** | **2** | **1** | **3** | **1** | **4** | **1** | **3** | **3** | **3** | **1** | **1** | **3** | **2** | **2** | **2** |
| Que. | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 |
| **Ans.** | **2** | **4** | **1** | **3** | **3** | **1** | **1** | **2** | **4** | **3** | **1** | **3** | **3** | **4** | **3** |
| Que. | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| **Ans.** | **4** | **1** | **1** | **2** | **1** | **2** | **1** | **3** | **1** | **1** | **3** | **4** | **4** | **3** | **2** |

## EXERCISE-III (Analytical Questions)

## Master Your Understanding

**1.** In the reaction $CH_3OH \rightarrow HCOOH$, the number of electrons that must be added to the right is :-

(1) 4 (2) 3 (3) 2 (4) 1

**RR0127**

**2.** Which statement is wrong :–

(1) Oxidation number of oxygen is +1 in peroxides

(2) Oxidation number of oxygen is +2 in oxygen difluoride

(3) Oxidation number of oxygen is $-\frac{1}{2}$ in superoxides

(4) Oxidation number of oxygen is –2 in most of its compound

**RR0128**

**3.** In the reaction $8Al + 3Fe_3O_4 \rightarrow 4Al_2O_3 + 9Fe$, the number of electrons transferred from reductant to oxidant is :-

(1) 8 (2) 4 (3) 16 (4) 24

**RR0129**

**4.** In which of the following reaction hydrogen is acting as an oxidising agent :-

(1) With iodine to give hydrogen iodide

(2) With lithium to give lithium hydride

(3) With nitrogen to give ammonia

(4) With sulphur to give hydrogen sulphide

**RR0130**

**5.** Oxidation number of Xe in $XeF_5^-$ is :

(1) +1 (2) +2 (3) +3 (4) +4

**RR0131**

**6.** Which is the best description of the behaviour of bromine in the reaction given below :-

$H_2O + Br_2 \rightarrow HOBr + HBr$

(1) Both oxidized and reduced

(2) Oxidized only

(3) Reduced only

(4) Proton acceptor only

**RR0132**

**7.** The correct order of acidic strength is –

(1) $HClO_4 < HClO_3 < HClO_2 < HClO$

(2) $HClO_2 < HClO_3 < HClO_4 < HClO$

(3) $HClO_4 < HClO < HClO_2 < HClO_3$

(4) $HClO < HClO_2 < HClO_3 < HClO_4$

**RR0133**

**8.** Consider the following reaction:

$$xMnO_4^- + yC_2O_4^{2-} + zH^+ \rightarrow xMn^{2+} + 2yCO_2 + \frac{z}{2} + H_2O$$

The values of x, y and z in the reaction are respectively:-

(1) 5,2 and 16 (2) 2,5 and 8

(3) 2, 5 and 16 (4) 5,2 and 8

**RR0134**

**9.** In which of the following reaction $H_2O_2$ acts as a reducing agent ?

(a) $H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$

(b) $H_2O_2 - 2e^- \rightarrow O_2 + 2H^+$

(c) $H_2O_2 + 2e^- \rightarrow 2OH^-$

(d) $H_2O_2 + 2OH^- - 2e^- \rightarrow O_2 + 2H_2O$

(1) (a), (c) (2) (b), (d) (3) (a), (b) (4) (c), (d)

**RR0135**

**10.** In the following reaction

$2FeCl_3 + H_2S \longrightarrow 2FeCl_2 + 2HCl + S$

(1) $FeCl_3$ is oxidant

(2) $FeCl_3$ & $H_2S$ are oxidised

(3) $FeCl_3$ is oxidised & $H_2S$ is reduced

(4) $H_2S$ is oxidant

**RR0137**

**11.** In oxidation reduction reaction

$Cr_2O_7^{-2} + SO_3^{-2} + H^+ \rightarrow Cr^{+3} + SO_4^{-2} + H_2O$ the balance equation coefficient for $Cr_2O_7^{-2}$, $SO_3^{-2}$, $H^+$ are :–

(1) 1, 3, 8 (2) 16, 5, 2

(3) 2, 16, 5 (4) 5, 2, 16

**RR0138**

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**12.** Which of the following act both as oxidant & reductant:–

(1) $H_2S$ (2) $SO_3$

(3) $H_2O_2$ (4) $F_2$

**RR0139**

**13.** Oxidation number of chlorine in perchloric acid is:-

(1) +1 (2) +3

(3) +5 (4) +7

**RR0141**

**14.** Identify the reducing agent in the reaction.

$3Fe_3O_4 + 8Al\ (s) \rightarrow 9Fe + 4Al_2O_3\ (s)$

(1) $Fe_3O_4$ (2) Al

(3) Fe (4) $Al_2O_3$

**RR0168**

**15.** Which of the following is a disproportionation reaction:

(1) $2KClO_3 \longrightarrow 2KCl + 3O_2$

(2) $CaCO_3 \longrightarrow CaO + CO_2$

(3) $2H_2O_2 \longrightarrow 2H_2O + O_2$

(4) $Fe^{+2} + MnO_4^- + 8H^+ \longrightarrow Fe^{+3} + Mn^{+2} + 4H_2O$

**RR0169**

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## EXERCISE-III (Analytical Questions) ANSWER KEY

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Ans.** | **1** | **1** | **4** | **2** | **4** | **1** | **4** | **3** | **2** | **1** | **1** | **3** | **4** | **2** | **3** |